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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -3

AIM: Launch an Ubuntu EC2 instance, install the Nginx web server through a user data script at launch time and open the default page in a browser. Record the differences you observed compared to the Amazon Linux and Apache steps.

Description:

In this experiment, an Ubuntu EC2 instance was launched successfully using the AWS Management Console. HTTP traffic was allowed through the security group to access the web server.

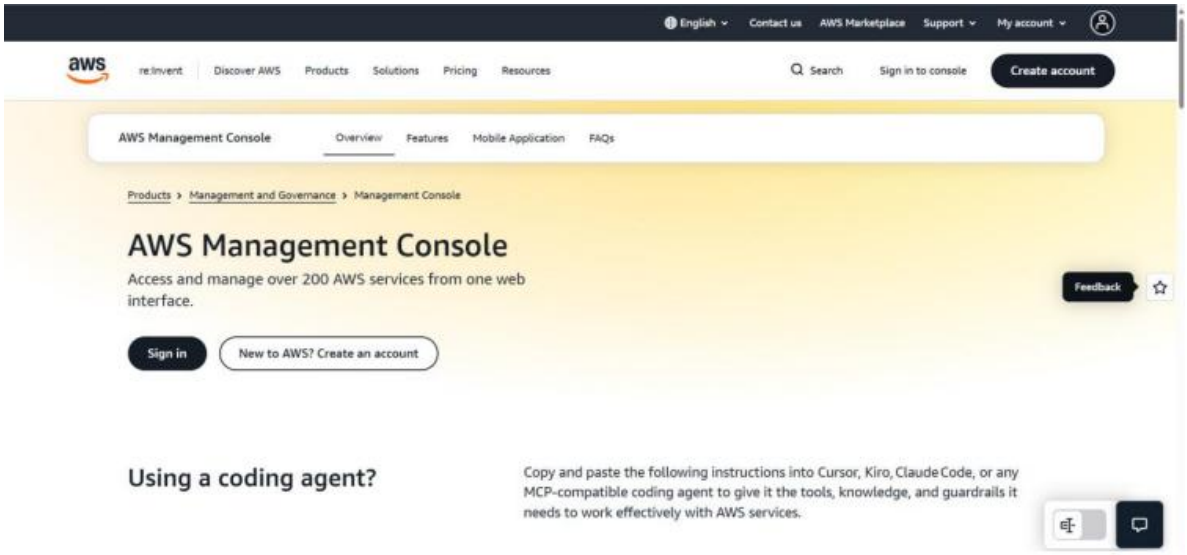
A User Data script was used to automatically update packages, install Nginx and start the web server during the first boot.

The Public IPv4 address was opened in a browser, and the default Nginx welcome page was displayed. Unlike Amazon Linux, Ubuntu uses apt for package management, and Nginx is used instead of Apache.

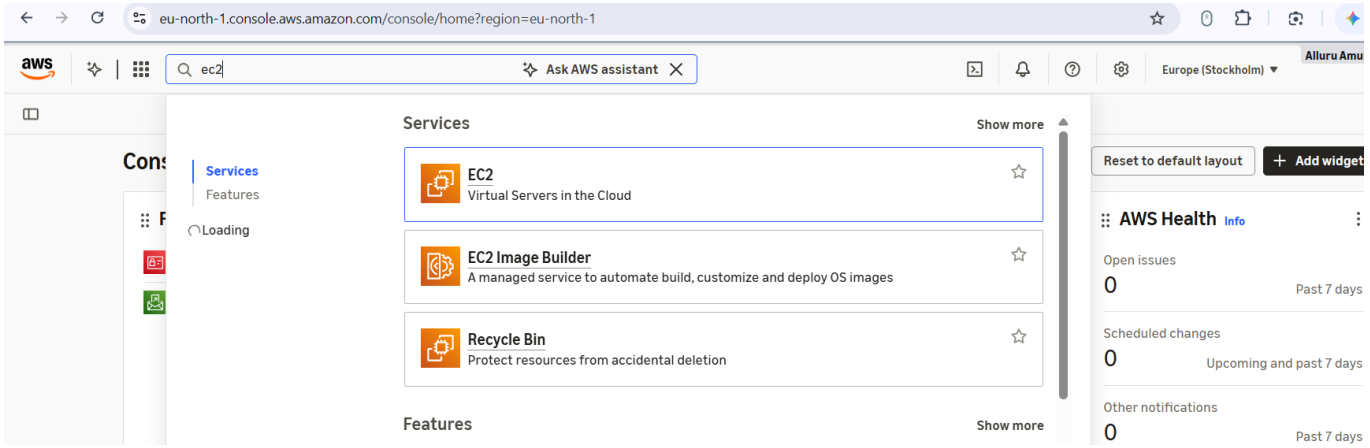
Steps:

Part A: Launch an Ubuntu EC2 Instance

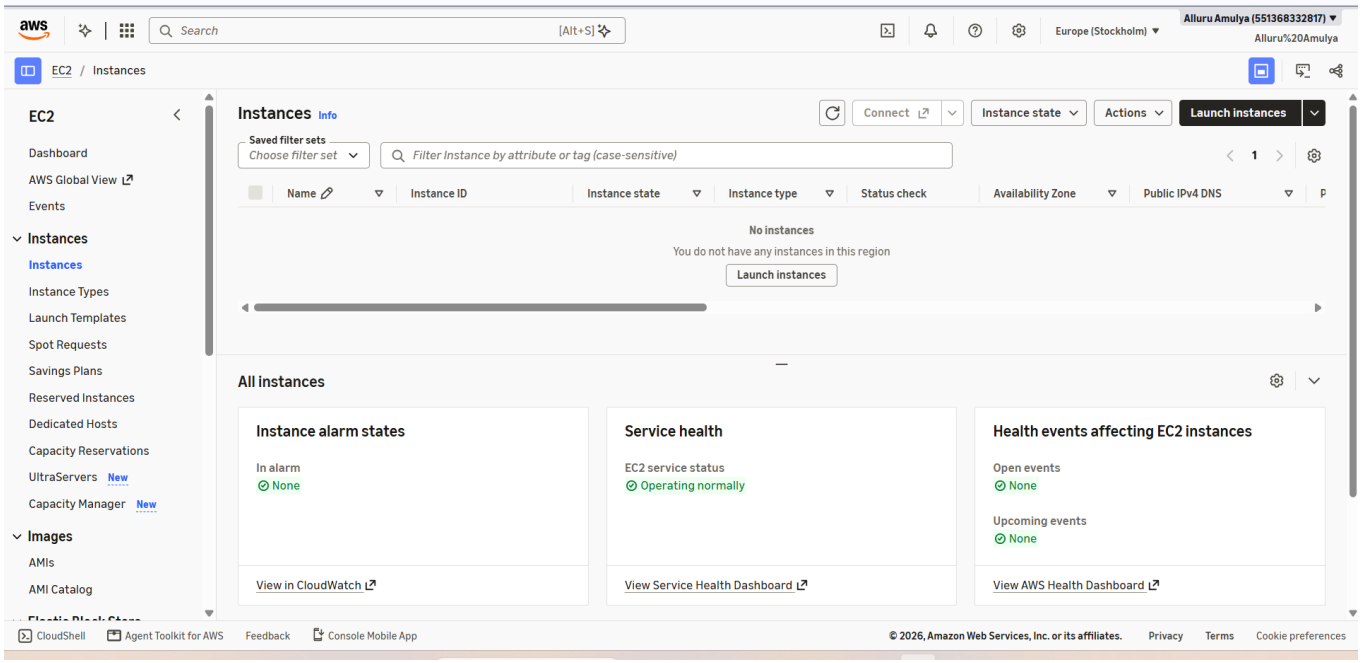
1. Log in to the AWS Management Console.



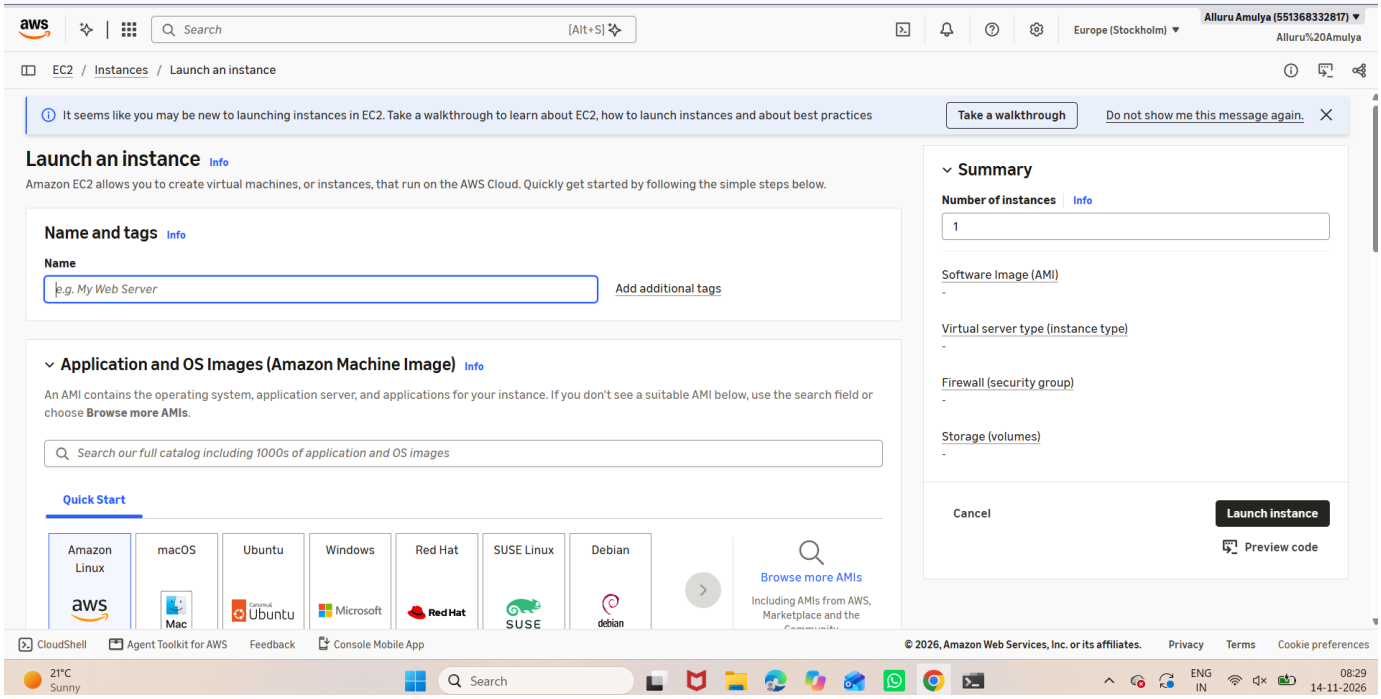
2. Search for EC2 in the AWS search bar and open the EC2 Dashboard.



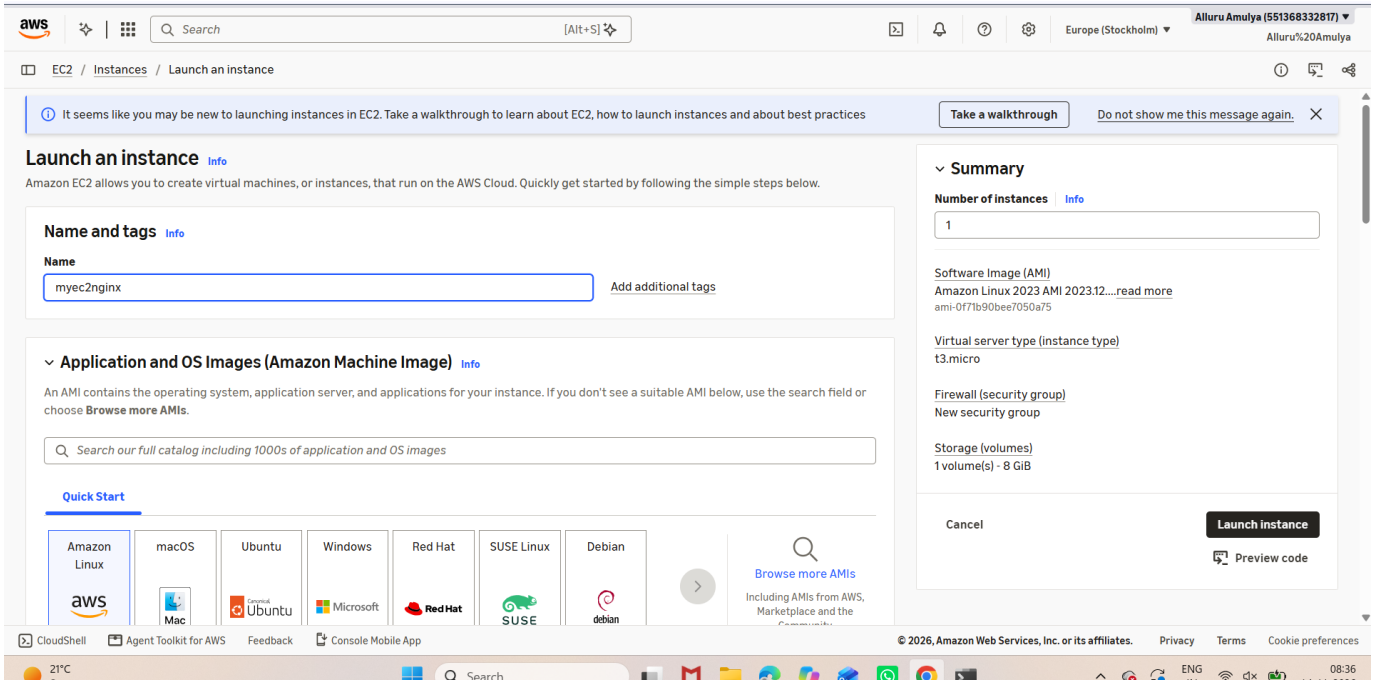
3. From the left navigation panel, select Instances.



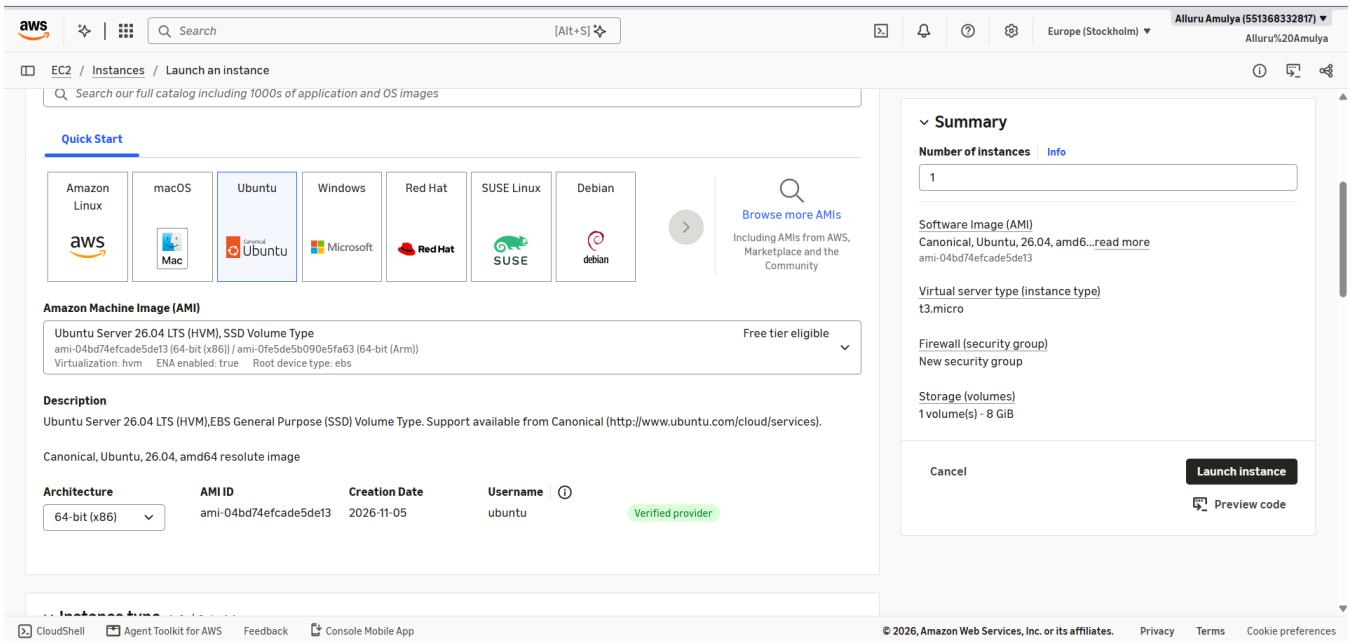
4. Click Launch instances.



5. Under Name and tags, enter an instance name (e.g., Ubuntu-Nginx-Server).



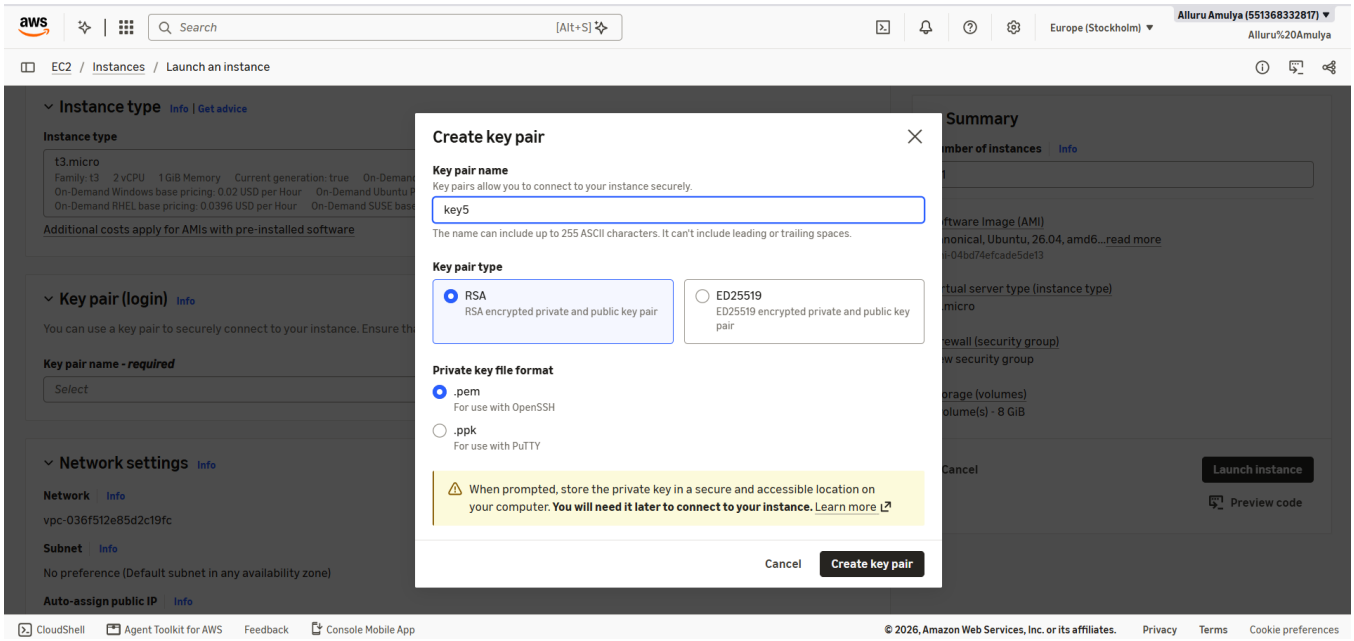
6. Under Application and OS Images (Amazon Machine Image), select Ubuntu.



7. Choose an Ubuntu Server LTS AMI (e.g., Ubuntu Server 26.04 LTS (HVM)).

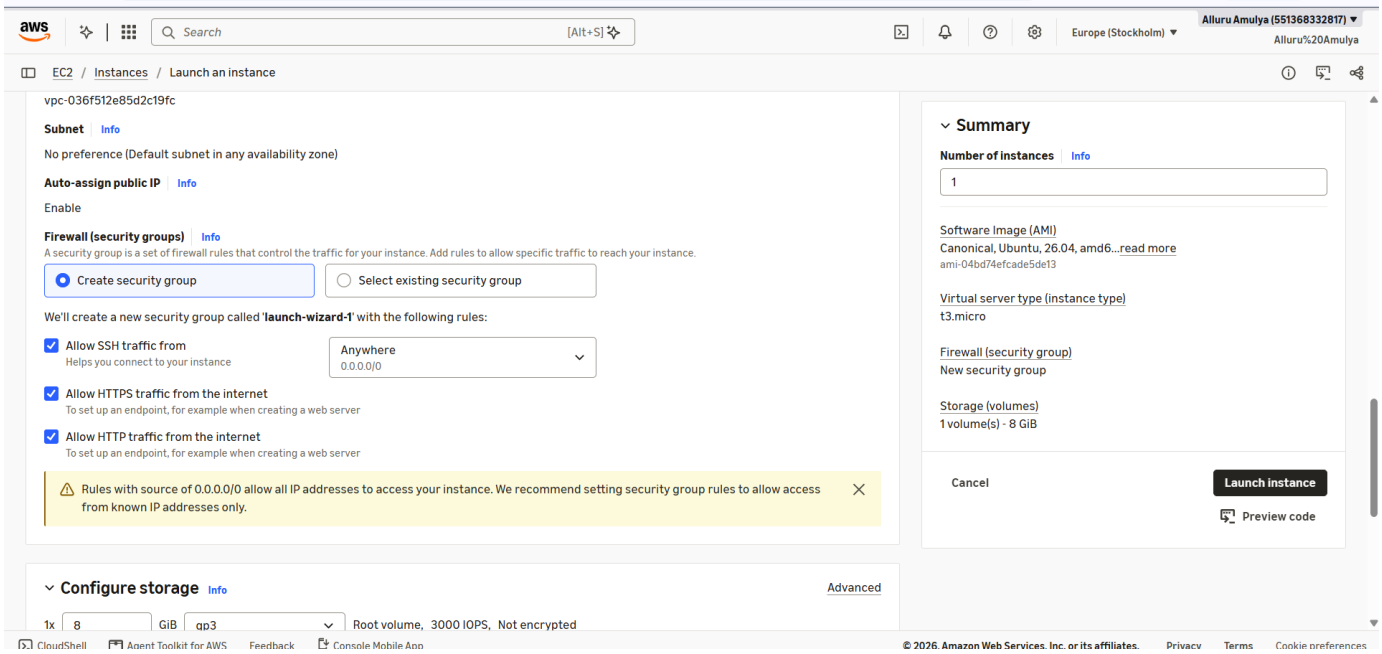
8. Under Instance type, select an eligible instance type, such as t3.micro, depending on availability and Free Tier eligibility.

9. Under Key pair (login), select an existing key pair or create a new key pair if SSH access is required.



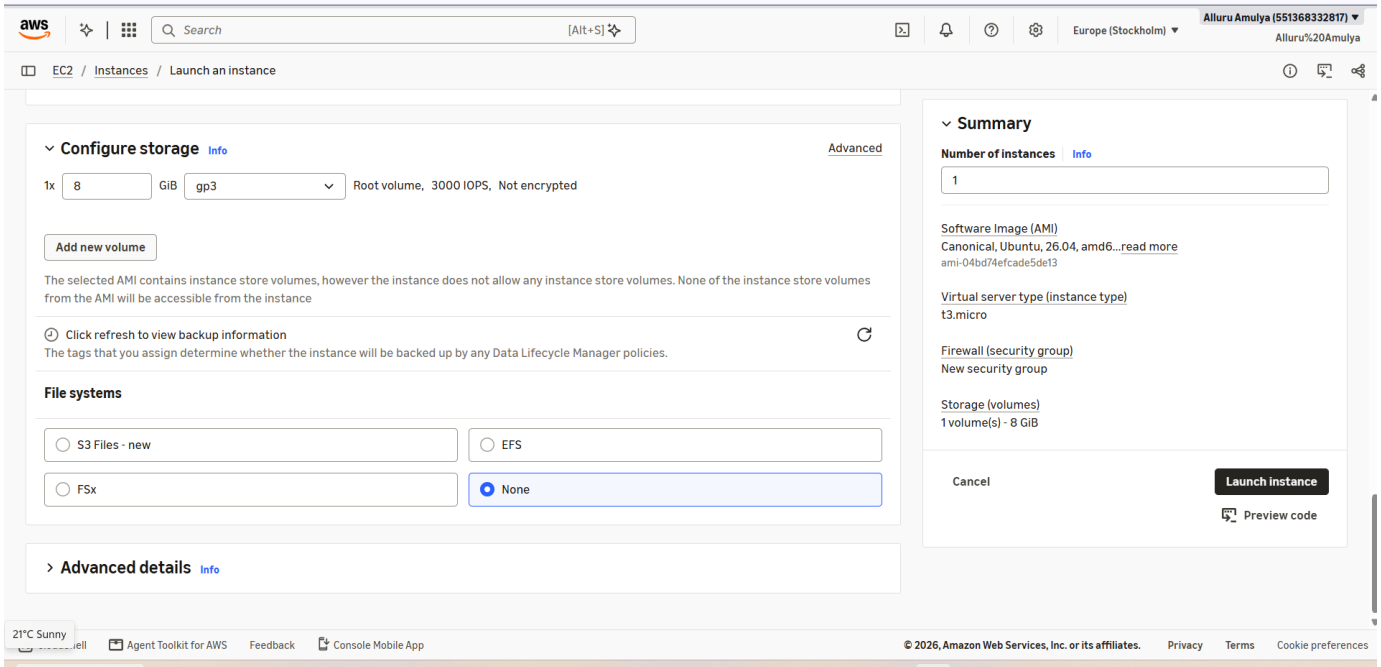
10. Under Network settings, configure the following:

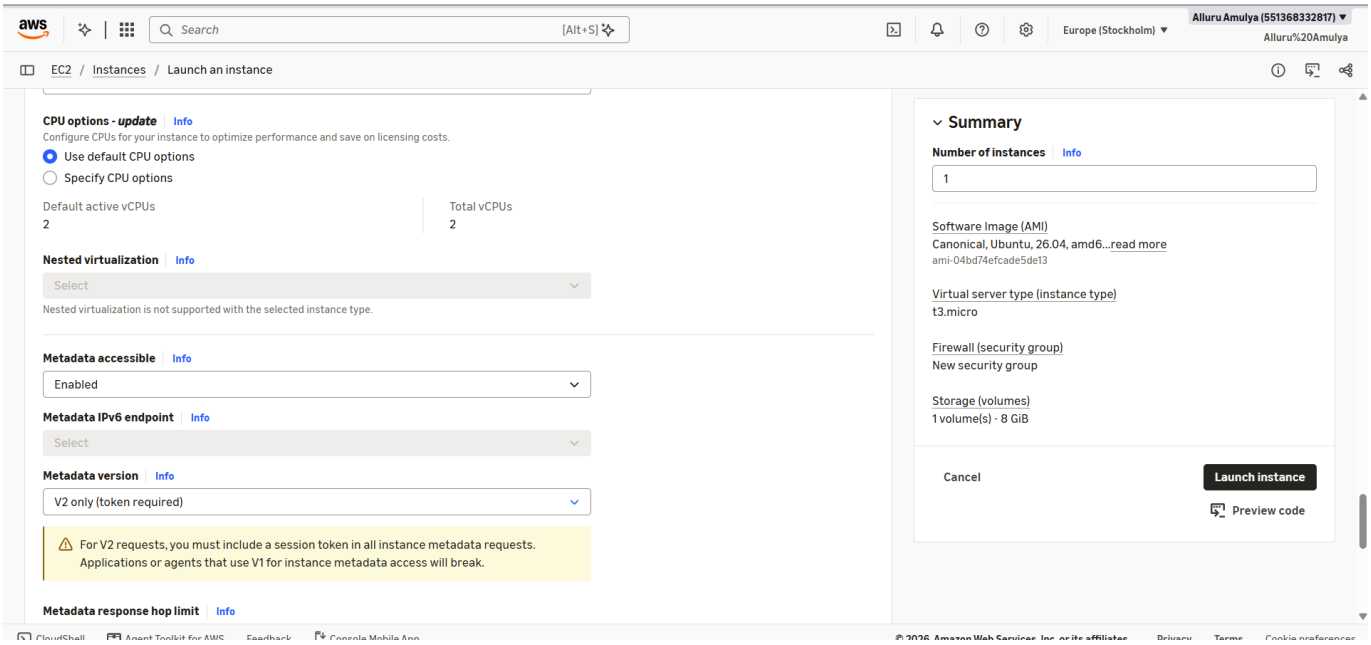
- Allow SSH traffic from.
- Allow HTTP traffic from the internet (3rd option)



11. Under Configure storage, retain the default storage settings unless additional storage is required.

12. Expand Advanced details. (Keep the default settings only)





13. Click Launch instance. (Present in bottom right)

Part B: Install Nginx Using the User Data Script

14. In the Advanced details section, scroll down to the User data text box.

15. Enter the following shell script to install and start the Nginx web server automatically when the instance launches.

User Data Script:

```
#!/bin/bash
# Update Ubuntu package repositories
sudo apt update -y

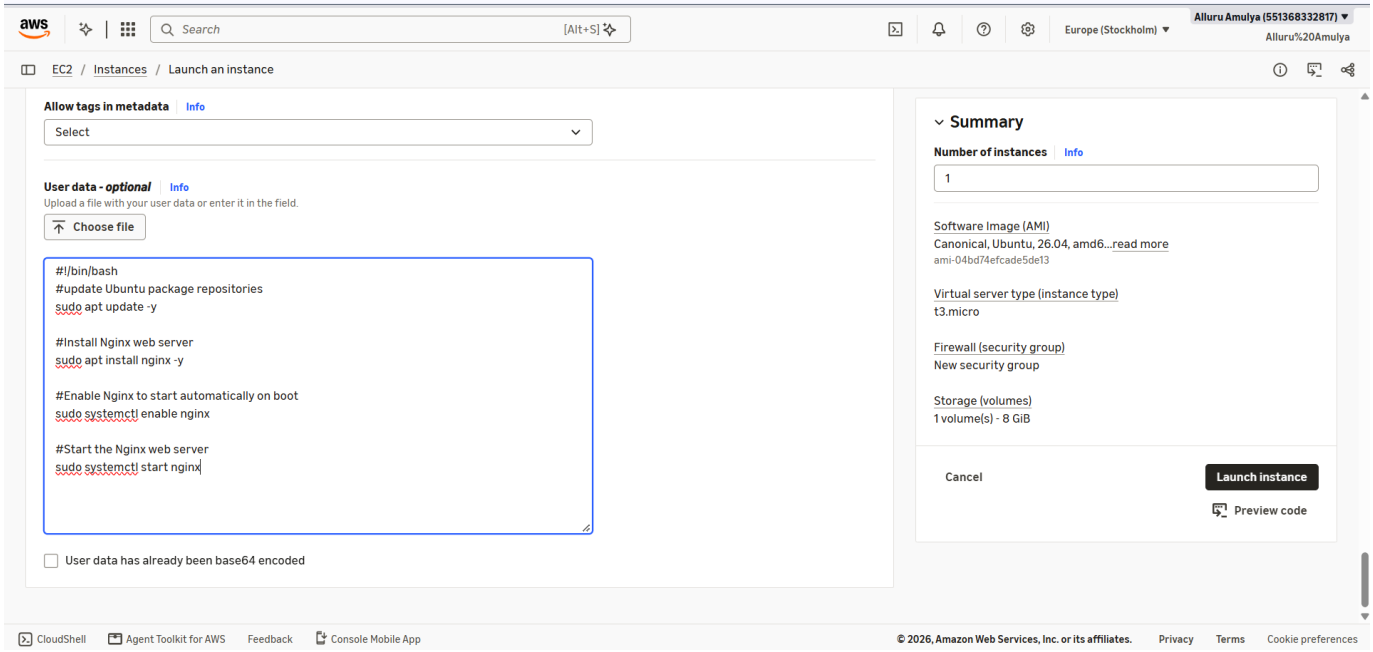
# Install Nginx web server
sudo apt install nginx -y
```

Enable Nginx to start automatically on boot

sudo systemctl enable nginx

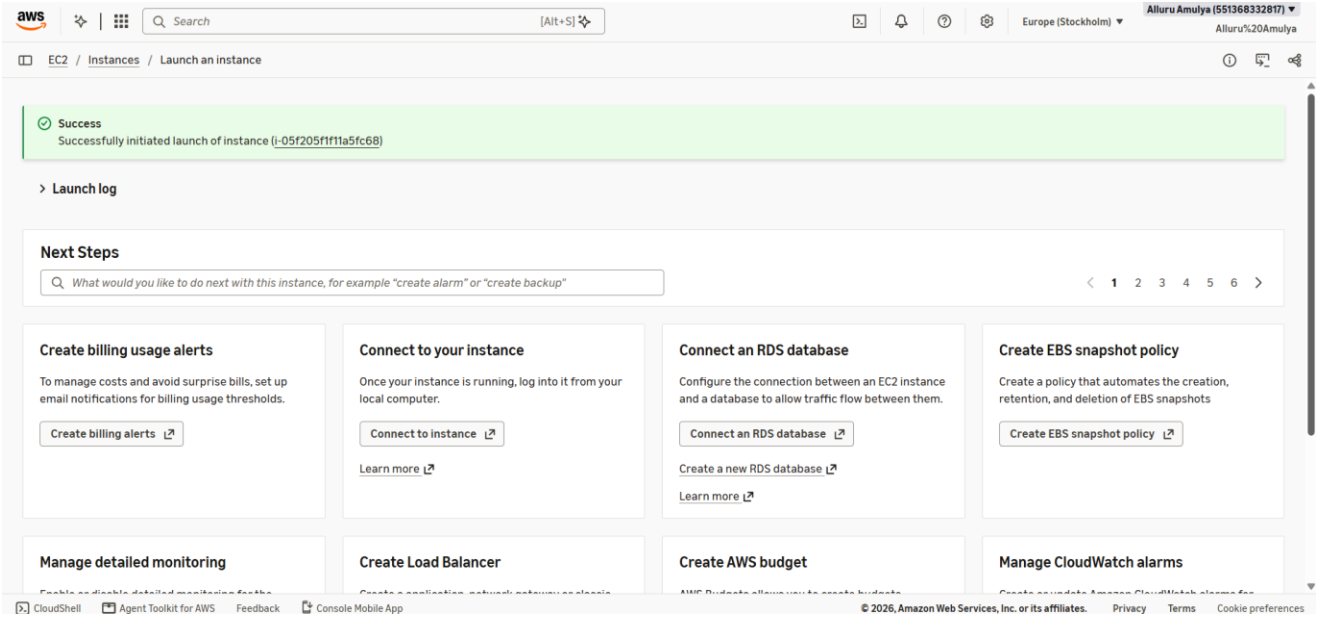
Start the Nginx web server

sudo systemctl start nginx



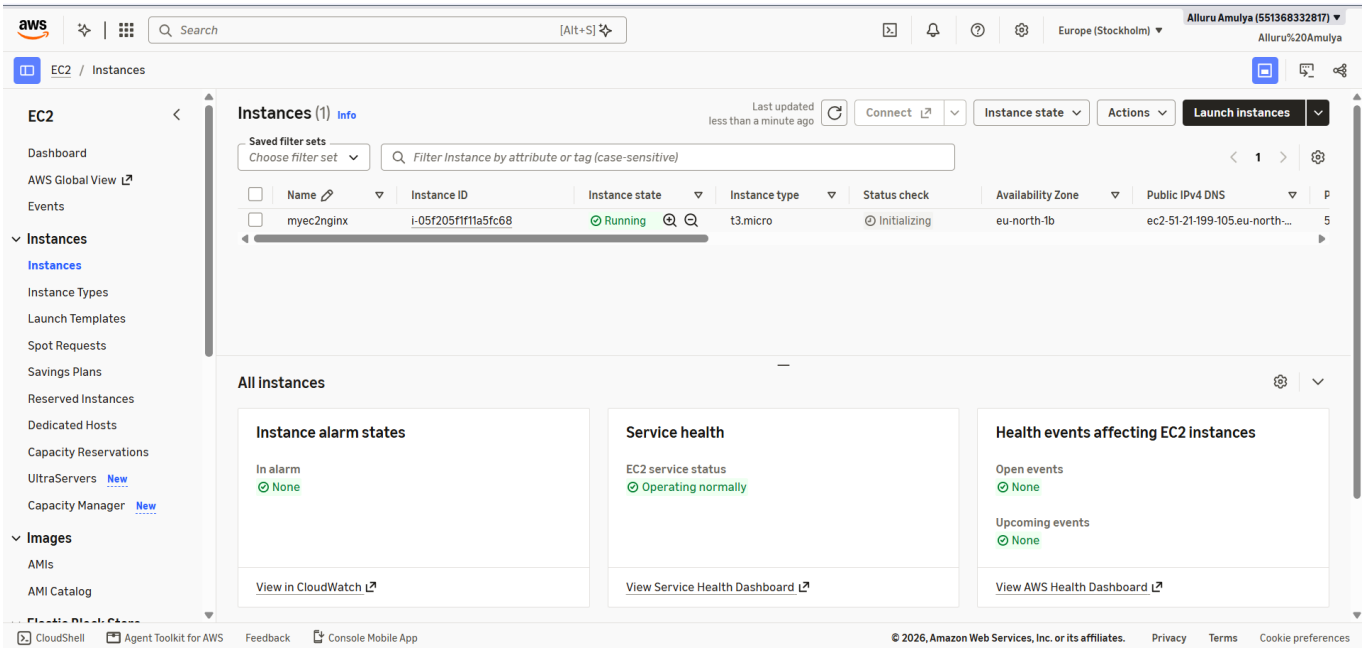
16. Verify that the script has been entered correctly in the User data field.

17. Click Launch instance.



18. Wait until the instance state changes to Running and the status checks show 2/2 checks passed.

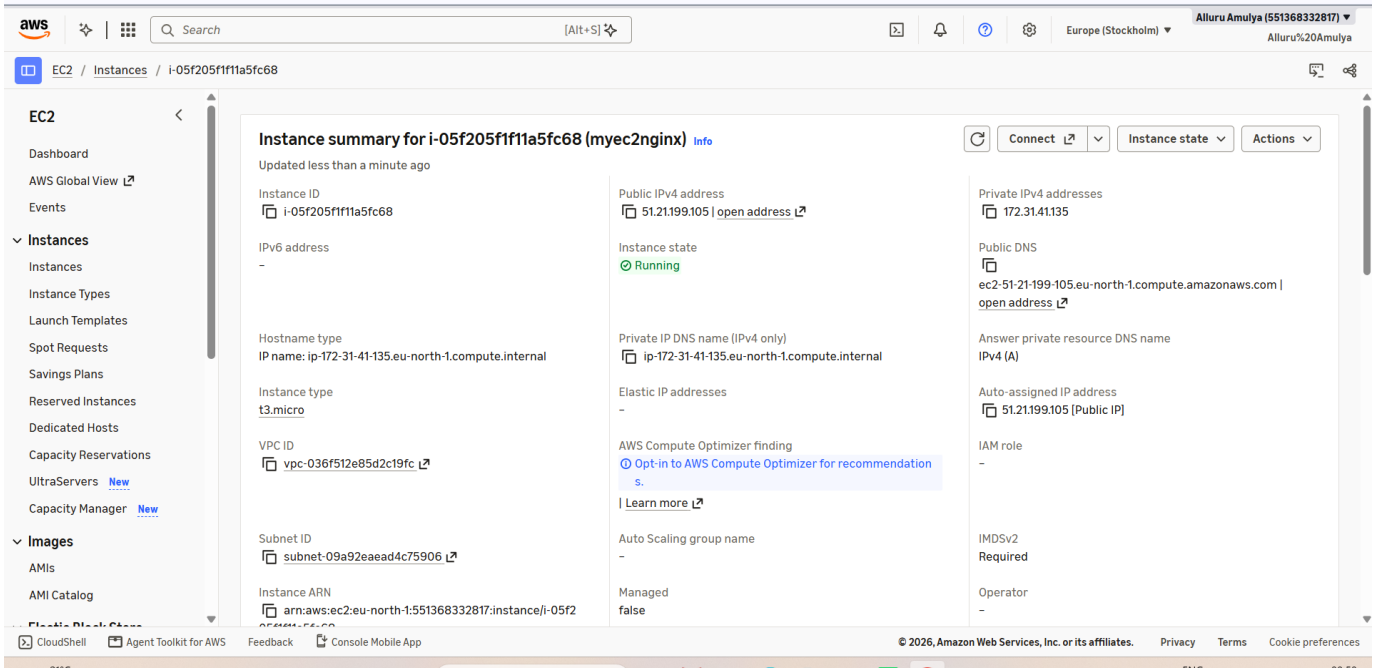
19. The User Data script executes automatically during the instance's first boot. It installs Nginx and starts the web server.



Important: User Data scripts for Linux instances normally execute during the first boot only. Allow a few minutes for package installation and server startup.

Part C: Access the Nginx Default Web Page

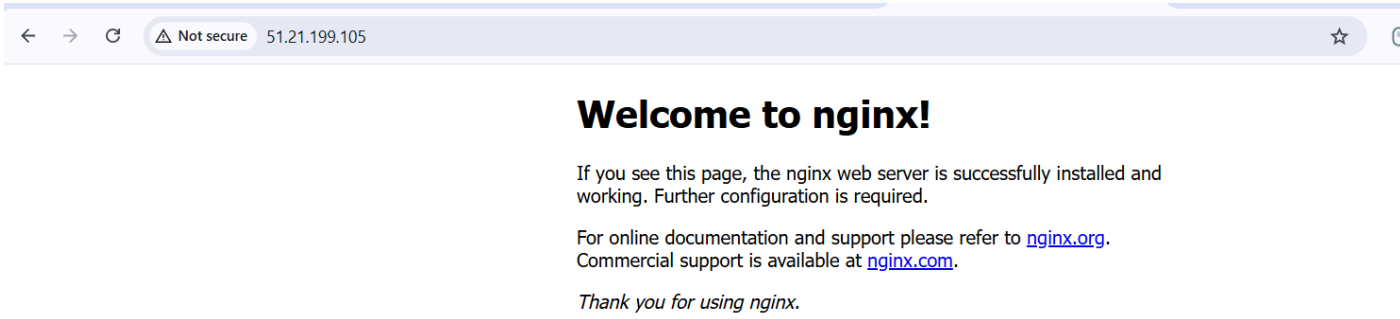
20. From the EC2 Dashboard, select the newly launched Ubuntu-Nginx-Server instance.



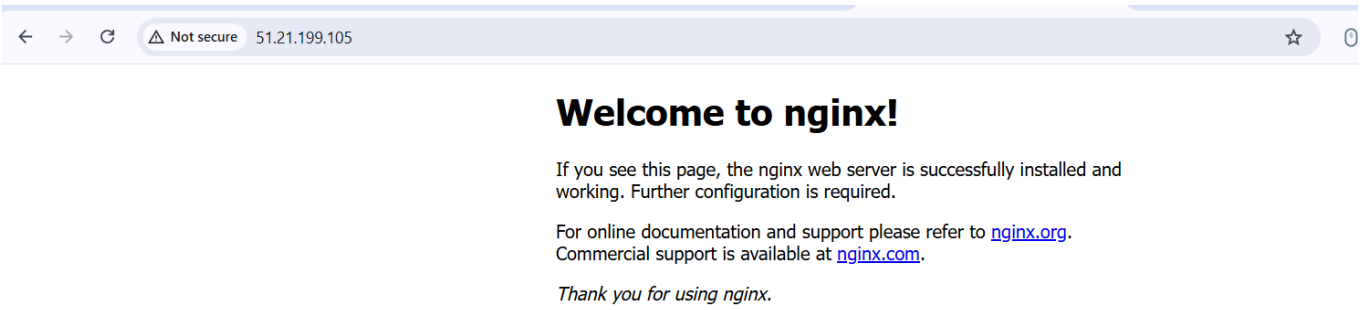
21. Locate and copy its Public IPv4 address.

22. Open a web browser.

23. Enter the following URL in the address bar:



Output:


Not secure
51.21.199.105
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.